

Building A Seismic Design

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1 Effective Seismic Weight Calculation

We calculate the effective seismic weight with unit weights and square footages.

1.1 Unit Weights

Weights will be assigned with area unit weights.

$$U_{corridor} = 0.035 \text{ ksf} \quad U_{floor} = 0.035 \text{ ksf} \quad U_{roof} = 0.025 \text{ ksf} \quad U_{wall} = 0.012 \text{ ksf}$$

1.2 Building Geometry

The lengths and widths of the weight areas are consistent across levels 2, 3, and 4.

$$L_{floor} = 230 \text{ ft} \quad w_{floor} = 30.75 \text{ ft} \quad L_{corridor} = 230 \text{ ft} \quad w_{corridor} = 5.5 \text{ ft} \quad w_{building} = 67 \text{ ft}$$

The wall heights vary across the levels.

$$h_{wall,1} = 11.5 \text{ ft} \quad h_{wall,2} = 10.5 \text{ ft} \quad h_{wall,3} = 10.5 \text{ ft} \quad h_{wall,4} = 10 \text{ ft}$$

There is a small jog in the building floorplan. This does not affect the area, but the perimeter will be longer.

$$L_{jog} = 5 \text{ ft}$$

$$\begin{aligned} P_{wall} &= (2 \cdot L_{floor}) + (2 \cdot w_{building}) + (2 \cdot L_{jog}) \\ &= (2 \cdot 230 \text{ ft}) + (2 \cdot 67 \text{ ft}) + (2 \cdot 5 \text{ ft}) \\ &= 604 \text{ ft} \end{aligned}$$

Areas can now be calculated. Note that the areas of the walls are the actual wall areas; they have not been distributed to each floor based on tributary area, yet.

$$\begin{aligned}
A_{\text{floor}} &= 2 \cdot (L_{\text{floor}} \cdot w_{\text{floor}}) = 2 \cdot (230 \text{ ft} \cdot 30.75 \text{ ft}) = 1.414e + 04 \text{ ft}^2 \\
A_{\text{corridor}} &= L_{\text{corridor}} \cdot w_{\text{corridor}} = 230 \text{ ft} \cdot 5.5 \text{ ft} = 1265 \text{ ft}^2 \\
A_{\text{level}} &= L_{\text{floor}} \cdot w_{\text{building}} = 230 \text{ ft} \cdot 67 \text{ ft} = 1.541e + 04 \text{ ft}^2 \\
A_{\text{wall},1} &= h_{\text{wall},1} \cdot P_{\text{wall}} = 11.5 \text{ ft} \cdot 604 \text{ ft} = 6946 \text{ ft}^2 \\
A_{\text{wall},2} &= h_{\text{wall},2} \cdot P_{\text{wall}} = 10.5 \text{ ft} \cdot 604 \text{ ft} = 6342 \text{ ft}^2 \\
A_{\text{wall},3} &= h_{\text{wall},3} \cdot P_{\text{wall}} = 10.5 \text{ ft} \cdot 604 \text{ ft} = 6342 \text{ ft}^2 \\
A_{\text{wall},4} &= h_{\text{wall},4} \cdot P_{\text{wall}} = 10 \text{ ft} \cdot 604 \text{ ft} = 6040 \text{ ft}^2
\end{aligned}$$

1.3 Component Weights

The typical floor and corridor weights are

$$\begin{aligned}
W_{\text{floor},\text{typ}} &= U_{\text{floor}} \cdot A_{\text{floor}} = 0.035 \text{ ksf} \cdot 1.414e + 04 \text{ ft}^2 = 495.1 \text{ kip} \\
W_{\text{corridor},\text{typ}} &= U_{\text{corridor}} \cdot A_{\text{corridor}} = 0.035 \text{ ksf} \cdot 1265 \text{ ft}^2 = 44.28 \text{ kip}
\end{aligned}$$

This applies to levels 2, 3, and 4.

$$\begin{aligned}
W_{\text{floor},2} &= 495.1 \text{ kip} & W_{\text{floor},3} &= 495.1 \text{ kip} & W_{\text{floor},4} &= 495.1 \text{ kip} \\
W_{\text{corridor},2} &= 44.28 \text{ kip} & W_{\text{corridor},3} &= 44.28 \text{ kip} & W_{\text{corridor},4} &= 44.28 \text{ kip}
\end{aligned}$$

The roof weight is

$$W_{\text{roof}} = U_{\text{roof}} \cdot A_{\text{level}} = 0.025 \text{ ksf} \cdot 1.541e + 04 \text{ ft}^2 = 385.2 \text{ kip}$$

The exterior walls are divided in half to the floor above and below.

$$\begin{aligned}
W_{\text{wall},2} &= U_{\text{wall}} \cdot \left(\frac{A_{\text{wall},2}}{2} + \frac{A_{\text{wall},1}}{2} \right) = 0.012 \text{ ksf} \cdot \left(\frac{6342 \text{ ft}^2}{2} + \frac{6946 \text{ ft}^2}{2} \right) = 79.73 \text{ kip} \\
W_{\text{wall},3} &= U_{\text{wall}} \cdot \left(\frac{A_{\text{wall},3}}{2} + \frac{A_{\text{wall},2}}{2} \right) = 0.012 \text{ ksf} \cdot \left(\frac{6342 \text{ ft}^2}{2} + \frac{6342 \text{ ft}^2}{2} \right) = 76.1 \text{ kip} \\
W_{\text{wall},4} &= U_{\text{wall}} \cdot \left(\frac{A_{\text{wall},4}}{2} + \frac{A_{\text{wall},3}}{2} \right) = 0.012 \text{ ksf} \cdot \left(\frac{6040 \text{ ft}^2}{2} + \frac{6342 \text{ ft}^2}{2} \right) = 74.29 \text{ kip} \\
W_{\text{wall},\text{roof}} &= U_{\text{wall}} \cdot \left(\frac{A_{\text{wall},4}}{2} \right) = 0.012 \text{ ksf} \cdot \left(\frac{6040 \text{ ft}^2}{2} \right) = 36.24 \text{ kip}
\end{aligned}$$

1.4 Level Weights

Each level's weight is

$$\begin{aligned}
W_{\text{level},2} &= W_{\text{floor},2} + W_{\text{corridor},2} + W_{\text{wall},2} = 495.1 \text{ kip} + 44.28 \text{ kip} + 79.73 \text{ kip} = 619.1 \text{ kip} \\
W_{\text{level},3} &= W_{\text{floor},3} + W_{\text{corridor},3} + W_{\text{wall},3} = 495.1 \text{ kip} + 44.28 \text{ kip} + 76.1 \text{ kip} = 615.5 \text{ kip} \\
W_{\text{level},4} &= W_{\text{floor},4} + W_{\text{corridor},4} + W_{\text{wall},4} = 495.1 \text{ kip} + 44.28 \text{ kip} + 74.29 \text{ kip} = 613.6 \text{ kip} \\
W_{\text{level},\text{roof}} &= W_{\text{roof}} + W_{\text{wall},\text{roof}} = 385.2 \text{ kip} + 36.24 \text{ kip} = 421.5 \text{ kip}
\end{aligned}$$

1.5 Total Weight

The total weight of the structure is

$$\begin{aligned}W_{total} &= W_{level,2} + W_{level,3} + W_{level,4} + W_{level,roof} \\&= 619.1 \text{ kip} + 615.5 \text{ kip} + 613.6 \text{ kip} + 421.5 \text{ kip} \\&= 2270 \text{ kip}\end{aligned}$$

2 Seismic Equivalent Lateral Force (ELF) Analysis

We will be using the equivalent lateral force method prescribed by ASCE 7-16 Section 12.8. We will envelope the forces with a fully flexible diaphragm assumption and a fully rigid diaphragm assumption.

2.1 Seismic Parameters

Seismic design parameters are obtained from the geotech report and ASCE 7-16 Sec. 11 and 12.

$$R = 6.5 \quad \Omega_0 = 3 \quad C_d = 4 \quad \rho = 1$$

$$\begin{aligned}I_e &= 1 & S_S &= 0.887 \text{ (g)} & S_1 &= 0.41 \text{ (g)} & F_a &= 1.145 & F_v &= 1.89 \\S_{MS} &= 1.016 \text{ (g)} & S_{M1} &= 0.7749 \text{ (g)} & S_{DS} &= 0.6772 \text{ (g)} & S_{D1} &= 0.5166 \text{ (g)} & SDC &= D \\C_u &= 1.4 & T_L &= 16 \text{ (s)}\end{aligned}$$

2.2 Base Shear

We must calculate the period

$$\begin{aligned}h_{top,roof} &= 57.67 \text{ ft} \\h_{bottom,roof} &= 45.5 \text{ ft} \\h_n &= 51.58 \text{ ft (ASCE 7-16 Sec. 11.2)} \\h_{n,check} &= 51.6 \text{ ft} \leq 65 \text{ ft} \quad \text{OK} \checkmark\end{aligned}$$

We will approximate the building period.

$$\begin{aligned}C_t &= 0.02 \text{ (ASCE 7-16 Table 12.8-2)} \\x &= 0.75 \text{ (ASCE 7-16 Table 12.8-2)} \\T_a &= C_t \cdot (h_n)^x = 0.02 \cdot (51.58)^{0.75} = 0.385 \text{ (ASCE 7-16 Eq. 12.8-7)}\end{aligned}$$

Because we are not using an automated analysis software, we can make the following assumption:

$$T = 0.385 \text{ (ASCE 7-16 Sec. 12.8.2)}$$

With the period, we can calculate the seismic response coefficient.

$$C_s = \frac{S_{DS}}{\frac{R}{I_e}} = \frac{0.6772}{\frac{6.5}{1}} = 0.1042 \text{ (ASCE 7-16 Eq. 12.8-2)}$$

We must check that C_s does not exceed:

Since, $T \leq T_L \rightarrow (0.385 \leq 16) : (ASCE 7-16 \text{ Sec. } 12.8.1.1)$

$$\begin{aligned} C_{s,check,max} &= \frac{S_{D1}}{T \cdot \left(\frac{R}{I_e}\right)} \\ &= \frac{0.5166}{0.385 \cdot \left(\frac{6.5}{1}\right)} \\ &= 0.104 \leq 0.206 \quad \text{OK} \checkmark \end{aligned}$$

$$C_{s,check,max} = 0.104 \leq 0.206 \quad \text{OK} \checkmark$$

And we must also check that C_s is not less than:

$$\begin{aligned} C_{s,check,min,1} &= 0.044 \cdot S_{DS} \cdot I_e = 0.044 \cdot 0.6772 \cdot 1 = 0.0298 \quad (ASCE 7-16 \text{ Eq. } 12.8-5) \\ C_{s,check,min} &= 0.104 \geq 0.0298 \quad \text{OK} \checkmark \end{aligned}$$

as well as:

$$C_{s,check,min} = 0.104 \geq 0.01 \quad \text{OK} \checkmark \quad (ASCE 7-16 \text{ Eq. } 12.8-5)$$

and finally:

Since, $S_1 < 0.6 \rightarrow (0.41 < 0.6) : (ASCE 7-16 \text{ Eq. } 12.8-6)$

$$C_{s,check,min} = \text{skipping check}$$

And now, we can calculate the base shear, V

$$V = C_s \cdot W_{total} = 0.1042 \cdot 2270 \text{ kip} = 236.5 \text{ kip} \quad (ASCE 7-16 \text{ Eq. } 12.8-1)$$

2.3 Distribute the seismic forces

With the base shear calculated, we must now distribute the base shear on each level.

First, we must calculate the structural period exponent, k .

Since, $T \leq 0.5 \rightarrow (0.385 \leq 0.5) :$

$$k = 1$$

Now, we can distribute the forces per floor,

$$F_x = C_{vx} V \quad (ASCE 7-16 \text{ Eq. } 12.8-11)$$

As well as cumulatively from roof to ground.

$$V_x = \sum_{i=x}^n F_i \quad (ASCE 7-16 \text{ Eq. } 12.8-11)$$

Calculating for each floor leads to the following [Table 1](#)

Table 1: Seismic Forces Calculation Table

Level	h_{floor} [ft]	w_x [kip]	h_x [ft]	$w_x h_x^k$	C_{vx}	F_x [kip]	V_x [kip]	OTM [kip-ft]
Roof	0.0	421	42.5	17913	0.306	72.4	0.0	0.0
Level 4	10.0	614	32.5	19943	0.341	80.6	72.4	724
Level 3	10.5	615	22.0	13540	0.231	54.7	153	2330
Level 2	10.5	619	11.5	7119	0.122	28.8	208	4511
Ground	11.5	0.0	0.0	0.0	0.0	0.0	236	7230

3 Diaphragm and Shear Wall Design

3.1 Redundancy Factor

The redundancy factor, ρ , is either 1.3 or 1.0 according to ASCE 7-16 12.3.4.

No single story resists more than 35% of the shear force (see C_{vx} in [Table 1](#)), so all values of ρ are 1.0.

Table: Redundancy Factors {#tab:redundancy_factors}

3.2 Envelope Procedure

We will envelope the forces with a fully flexible diaphragm assumption and a fully rigid diaphragm assumption.

3.2.1 Flexible Diaphragm Assumption

We will uniformly distribute the shear forces across the diaphragms based on tributary areas of each level. This removes some nuance in the distribution of floor weight distribution (corridors are 17% heavier than the rest of the floors, for instance), but the vast majority of the area of each level is typical floor weight (30 psf) instead of corridor weight (35 psf).

$$f_{p,x} = F_x / A_{level}$$

$$A_{level} = 1.541e + 04 \text{ ft}^2$$

$$f_{p,roof} = 4.697 \text{ psf}$$

$$f_{p,4} = 5.23 \text{ psf}$$

$$f_{p,3} = 3.551 \text{ psf}$$

$$f_{p,2} = 1.867 \text{ psf}$$

North-South Direction Most of the shear walls are not in line which presents an issue: if we assume a diaphragm is the full depth, there will be an extremely long collector element without a shear wall on both sides. We will avoid this by just assuming the flexible diaphragm does not extend the full depth and instead is just tributary to the nearest wall ([Figure 1](#)). This means that there are essentially two diaphragms

through the depth of the building. This is a rough approximation to ensure all the shear load is resolved in the shear walls.

The walls that are close to inline (A-5 and A-5.1, A-6 and A-6.1, and A-9 and A-9.1) in reality will be treated as inline and have collectors to transfer the shear load from the diaphragm to those walls.

Proper detailing will be done to ensure continuity through the depth of the diaphragm.

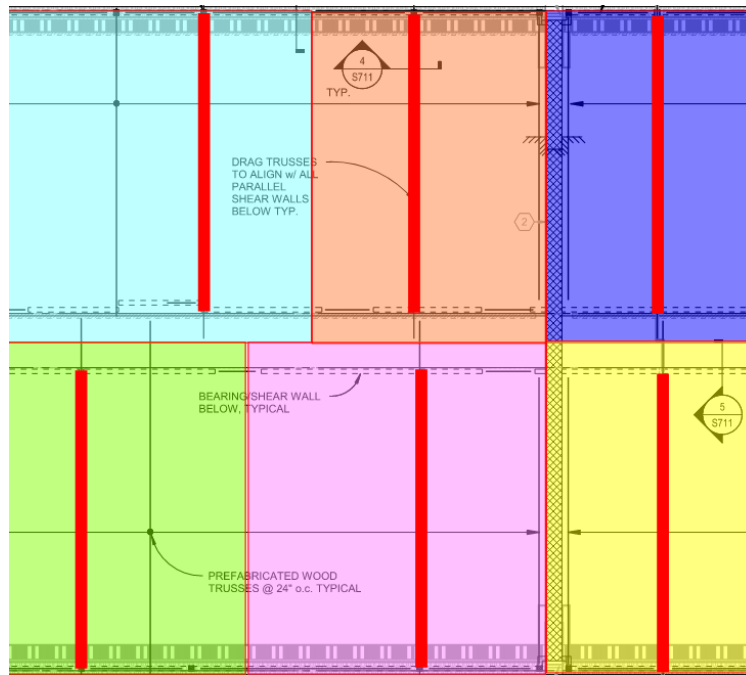


Figure 1: Shear walls (thick red lines) in plan view with their (colored) tributary areas.

Shear wall shear forces are tabulated in [Table 2](#), [Table 3](#), [Table 4](#), and [Table 5](#).

Shear wall sheathing will be assumed to have the following properties:

1. WSP Sheathing
2. 15/32" thickness
3. 10d common nails
4. Plywood panels
5. Framing uses Douglas-Fir-Larch (no specific gravity adjustment)

Notes:

- Cumulative shear force, $F_{x,cum}$, is cumulative from the Roof downwards. For example, Level 3 includes the shear forces from the Roof, Level 4, and Level 3.

Check aspect ratios We must check the aspect ratios of the shear walls to ensure we don't need to apply the aspect ratio factor in SDPWS Sec. 4.3.3.2. The maximum aspect ratio of all N-S walls is:

$$\text{ratio}_{max} = 0.575 < 2 \quad \text{OK} \checkmark$$

∴ All aspect ratios are OK

Table 2: Shear wall forces NS - Roof

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	4.70	1662	7809	7809	30.8	254	178	311	1.00	6.00
A-2.4	4.70	2147	10084	10084	30.8	328	230	311	1.00	6.00
A-3.7	4.70	741	3482	3482	20.0	174	122	311	1.00	6.00
A-4	4.70	505	2374	2374	20.0	119	83.1	311	1.00	6.00
A-5	4.70	771	3619	3619	30.8	118	82.4	311	1.00	6.00
A-5.1	4.70	1385	6504	6504	30.8	212	148	311	1.00	6.00
A-6	4.70	988	4642	4642	30.8	151	106	311	1.00	6.00
A-6.1	4.70	826	3882	3882	30.8	126	88.4	311	1.00	6.00
A-6.8	4.70	906	4255	4255	30.8	138	96.9	311	1.00	6.00
A-7	4.70	1012	4754	4754	30.8	155	108	311	1.00	6.00
A-7.8	4.70	641	3010	3010	20.0	150	105	311	1.00	6.00
A-8	4.70	794	3731	3731	30.8	121	84.9	311	1.00	6.00
A-8.2	4.70	448	2105	2105	20.0	105	73.7	311	1.00	6.00
A-8.9	4.70	1353	6354	6354	30.8	207	145	311	1.00	6.00
A-9	4.70	1231	5783	5783	30.8	188	132	311	1.00	6.00

Table 3: Shear wall forces NS - Level 4

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	5.23	1662	8694	16503	30.8	537	376	461	1.00	4.00
A-2.4	5.23	2147	11227	21312	30.8	693	485	600	1.00	3.00
A-3.7	5.23	741	3876	7358	20.0	368	258	311	1.00	6.00
A-4	5.23	505	2643	5016	20.0	251	176	311	1.00	6.00
A-5	5.23	771	4030	7649	30.8	249	174	311	1.00	6.00
A-5.1	5.23	1385	7241	13746	30.8	447	313	461	1.00	4.00
A-6	5.23	988	5168	9811	30.8	319	223	311	1.00	6.00
A-6.1	5.23	826	4322	8203	30.8	267	187	311	1.00	6.00
A-6.8	5.23	906	4738	8993	30.8	292	205	311	1.00	6.00
A-7	5.23	1012	5292	10046	30.8	327	229	311	1.00	6.00
A-7.8	5.23	641	3351	6360	20.0	318	223	311	1.00	6.00
A-8	5.23	794	4154	7884	30.8	256	179	311	1.00	6.00
A-8.2	5.23	448	2343	4448	20.0	222	156	311	1.00	6.00
A-8.9	5.23	1353	7074	13427	30.8	437	306	311	1.00	6.00
A-9	5.23	1231	6438	12222	30.8	397	278	311	1.00	6.00

Table 4: Shear wall forces NS - Level 3

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	3.55	1662	5903	22406	30.8	729	510	600	1.00	3.00
A-2.4	3.55	2147	7622	28934	30.8	941	659	770	1.00	2.00
A-3.7	3.55	741	2632	9990	20.0	499	350	461	1.00	4.00
A-4	3.55	505	1794	6810	20.0	341	238	311	1.00	6.00
A-5	3.55	771	2736	10385	30.8	338	236	311	1.00	6.00
A-5.1	3.55	1385	4916	18662	30.8	607	425	461	1.00	4.00
A-6	3.55	988	3509	13319	30.8	433	303	311	1.00	6.00
A-6.1	3.55	826	2934	11137	30.8	362	254	311	1.00	6.00
A-6.8	3.55	906	3216	12209	30.8	397	278	311	1.00	6.00
A-7	3.55	1012	3593	13639	30.8	444	310	311	1.00	6.00
A-7.8	3.55	641	2275	8635	20.0	432	302	311	1.00	6.00
A-8	3.55	794	2820	10704	30.8	348	244	311	1.00	6.00
A-8.2	3.55	448	1591	6039	20.0	302	211	311	1.00	6.00
A-8.9	3.55	1353	4802	18230	30.8	593	415	461	1.00	4.00
A-9	3.55	1231	4371	16593	30.8	540	378	461	1.00	4.00

Table 5: Shear wall forces NS - Level 2

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	1.87	1662	3104	25510	30.8	830	581	600	1.00	3.00
A-2.4	1.87	2147	4008	32942	30.8	1071	750	770	1.00	2.00
A-3.7	1.87	741	1384	11373	20.0	569	398	461	1.00	4.00
A-4	1.87	505	943	7754	20.0	388	271	311	1.00	6.00
A-5	1.87	771	1438	11823	30.8	384	269	311	1.00	6.00
A-5.1	1.87	1385	2585	21247	30.8	691	484	600	1.00	3.00
A-6	1.87	988	1845	15164	30.8	493	345	461	1.00	4.00
A-6.1	1.87	826	1543	12680	30.8	412	289	311	1.00	6.00
A-6.8	1.87	906	1691	13901	30.8	452	316	461	1.00	4.00
A-7	1.87	1012	1889	15529	30.8	505	353	461	1.00	4.00
A-7.8	1.87	641	1196	9831	20.0	492	344	461	1.00	4.00
A-8	1.87	794	1483	12187	30.8	396	277	311	1.00	6.00
A-8.2	1.87	448	836	6875	20.0	344	241	311	1.00	6.00
A-8.9	1.87	1353	2525	20755	30.8	675	472	600	1.00	3.00
A-9	1.87	1231	2298	18891	30.8	614	430	461	1.00	4.00

East-West Direction Most likely, the shear walls in the E-W direction will be governed by a rigid diaphragm assumption.

Shear wall shear forces are tabulated in [Table 6](#), [Table 7](#), [Table 8](#), and [Table 9](#).

Check aspect ratios We will also check the aspect ratios of the E-W walls.

$$\text{ratio}_{\max} = 1.7 < 2 \quad \text{OK} \checkmark$$

∴ All aspect ratios are OK

Table 6: Shear wall forces EW - Roof

Wall	W_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [plf]	v_{cap} [plf]	sides	s_{nail} [in]
A-B.1	4.70	1131	5311	5311	21.0	253	177	311	1.00	6.00
A-B.2.5	4.70	745	3501	3501	13.2	264	185	311	1.00	6.00
A-B.3.5	4.70	536	2518	2518	9.00	280	196	311	1.00	6.00
A-B.4	4.70	636	2990	2990	12.0	249	174	311	1.00	6.00
A-B.4.8	4.70	536	2518	2518	11.0	229	160	311	1.00	6.00
A-B.5.5	4.70	1156	5429	5429	29.8	182	128	311	1.00	6.00
A-B.6.5	4.70	946	4446	4446	12.8	349	244	311	1.00	6.00
A-B.7.9	4.70	955	4485	4485	16.2	276	193	311	1.00	6.00
A-B.8.8	4.70	1064	4996	4996	13.8	363	254	311	1.00	6.00
A-C.1.8	4.70	1474	6924	6924	17.2	401	281	311	1.00	6.00
A-C.3.2	4.70	1038	4878	4878	23.0	212	148	311	1.00	6.00
A-C.4.2	4.70	921	4328	4328	22.5	192	135	311	1.00	6.00
A-C.5.5	4.70	1038	4878	4878	26.0	188	131	311	1.00	6.00
A-C.6.5	4.70	519	2439	2439	11.2	217	152	311	1.00	6.00
A-C.6.9	4.70	413	1942	1942	7.00	277	194	311	1.00	6.00
A-C.7.5	4.70	636	2990	2990	6.75	443	310	311	1.00	6.00
A-C.8.5	4.70	715	3357	3357	10.2	327	229	311	1.00	6.00
A-C.9.2	4.70	949	4458	4458	25.2	177	124	311	1.00	6.00

Table 7: Shear wall forces EW - Level 4

Wall	W_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [plf]	v_{cap} [plf]	sides	s_{nail} [in]
A-B.1	5.23	1131	5913	11224	21.0	534	374	461	1.00	4.00
A-B.2.5	5.23	745	3898	7399	13.2	558	391	461	1.00	4.00
A-B.3.5	5.23	536	2803	5321	9.00	591	414	461	1.00	4.00
A-B.4	5.23	636	3329	6319	12.0	527	369	461	1.00	4.00
A-B.4.8	5.23	536	2803	5321	11.0	484	339	461	1.00	4.00
A-B.5.5	5.23	1156	6044	11473	29.8	386	270	311	1.00	6.00
A-B.6.5	5.23	946	4949	9395	12.8	737	516	600	1.00	3.00
A-B.7.9	5.23	955	4993	9478	16.2	583	408	461	1.00	4.00
A-B.8.8	5.23	1064	5563	10559	13.8	768	538	600	1.00	3.00
A-C.1.8	5.23	1474	7709	14633	17.2	848	594	600	1.00	3.00
A-C.3.2	5.23	1038	5431	10309	23.0	448	314	461	1.00	4.00
A-C.4.2	5.23	921	4818	9145	22.5	406	285	311	1.00	6.00
A-C.5.5	5.23	1038	5431	10309	26.0	397	278	311	1.00	6.00
A-C.6.5	5.23	519	2716	5155	11.2	458	321	461	1.00	4.00
A-C.6.9	5.23	413	2162	4104	7.00	586	410	461	1.00	4.00
A-C.7.5	5.23	636	3329	6319	6.75	936	655	770	1.00	2.00
A-C.8.5	5.23	715	3737	7094	10.2	692	484	600	1.00	3.00
A-C.9.2	5.23	949	4963	9421	25.2	373	261	311	1.00	6.00

Table 8: Shear wall forces EW - Level 3

Wall	W_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [plf]	v_{cap} [plf]	sides	s_{nail} [in]
A-B.1	3.55	1131	4014	15238	21.0	726	508	600	1.00	3.00
A-B.2.5	3.55	745	2647	10046	13.2	758	531	600	1.00	3.00
A-B.3.5	3.55	536	1903	7224	9.00	803	562	600	1.00	3.00
A-B.4	3.55	636	2260	8579	12.0	715	500	600	1.00	3.00
A-B.4.8	3.55	536	1903	7224	11.0	657	460	461	1.00	4.00
A-B.5.5	3.55	1156	4104	15577	29.8	524	367	461	1.00	4.00
A-B.6.5	3.55	946	3360	12755	12.8	1000	700	770	1.00	2.00
A-B.7.9	3.55	955	3390	12868	16.2	792	554	600	1.00	3.00
A-B.8.8	3.55	1064	3777	14335	13.8	1043	730	770	1.00	2.00
A-C.1.8	3.55	1474	5234	19866	17.2	1152	806	921	2.00	4.00
A-C.3.2	3.55	1038	3687	13997	23.0	609	426	461	1.00	4.00
A-C.4.2	3.55	921	3271	12416	22.5	552	386	461	1.00	4.00
A-C.5.5	3.55	1038	3687	13997	26.0	538	377	461	1.00	4.00
A-C.6.5	3.55	519	1844	6998	11.2	622	435	461	1.00	4.00
A-C.6.9	3.55	413	1468	5572	7.00	796	557	600	1.00	3.00
A-C.7.5	3.55	636	2260	8579	6.75	1271	890	921	2.00	4.00
A-C.8.5	3.55	715	2537	9631	10.2	940	658	770	1.00	2.00
A-C.9.2	3.55	949	3370	12791	25.2	507	355	461	1.00	4.00

Table 9: Shear wall forces EW - Level 2

Wall	W_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [plf]	v_{cap} [plf]	sides	s_{nail} [in]
A-B.1	1.87	1131	2111	17349	21.0	826	578	600	1.00	3.00
A-B.2.5	1.87	745	1392	11438	13.2	863	604	770	1.00	2.00
A-B.3.5	1.87	536	1001	8225	9.00	914	640	770	1.00	2.00
A-B.4	1.87	636	1188	9767	12.0	814	570	600	1.00	3.00
A-B.4.8	1.87	536	1001	8225	11.0	748	523	600	1.00	3.00
A-B.5.5	1.87	1156	2158	17735	29.8	596	417	461	1.00	4.00
A-B.6.5	1.87	946	1767	14522	12.8	1139	797	921	2.00	4.00
A-B.7.9	1.87	955	1782	14650	16.2	902	631	770	1.00	2.00
A-B.8.8	1.87	1064	1986	16321	13.8	1187	831	921	2.00	4.00
A-C.1.8	1.87	1474	2752	22618	17.2	1311	918	921	2.00	4.00
A-C.3.2	1.87	1038	1939	15936	23.0	693	485	600	1.00	3.00
A-C.4.2	1.87	921	1720	14136	22.5	628	440	461	1.00	4.00
A-C.5.5	1.87	1038	1939	15936	26.0	613	429	461	1.00	4.00
A-C.6.5	1.87	519	969	7968	11.2	708	496	600	1.00	3.00
A-C.6.9	1.87	413	772	6343	7.00	906	634	770	1.00	2.00
A-C.7.5	1.87	636	1188	9767	6.75	1447	1013	1200	2.00	3.00
A-C.8.5	1.87	715	1334	10965	10.2	1070	749	770	1.00	2.00
A-C.9.2	1.87	949	1772	14563	25.2	577	404	461	1.00	4.00

3.2.2 Rigid Diaphragm Assumption

With a rigid diaphragm assumption, we now must calculate the following in each direction:

1. The relative stiffnesses of the shear walls to obtain a center of rigidity.
2. The center of mass of the diaphragm with the center of rigidity to obtain an eccentricity.

The relative stiffnesses of the shear walls can be calculated by using the deflection equation from NDS SDPWS Sec. 4.3.4

$$k = \frac{F}{\delta}, \text{ assume } F = \text{induced load demand in the shear wall}$$

$$\delta_{sw} = \frac{8vh^3}{EAb} + \frac{vh}{1000G_a} + \frac{h\Delta_A}{b} \quad (\text{SDPWS Eq. 4.3-1})$$

where

- b = the shear wall length [ft]
- Δ_A = vertical deformation of the wall overturning anchorage system + vertical compression deformation [in]
- E = modulus of elasticity of end posts [psi]
- A = area of end post cross-section [in²]
- G_a = apparent shear wall shear stiffness from nail slip and panel shear deformation [kip/in]
- h = shear wall height [ft]
- v = unit shear force induced by the design load [plf]
- δ_{sw} = maximum shear wall deflection determined by elastic analysis [in]

Note that for the following calculations, we will conservatively use the unit shear force induced by the nominal load, not the design load, for the value of v . This is because the stiffnesses obtained from this use will allow us to use the stiffnesses for deflection and story drift calculations

We will assume that the end posts are (2) 2x4 Douglas Fir No. 2; this means:

$$E_{endpost} = 1.6e + 06 \text{ psi}$$

$$A_{endpost} = 2 \cdot 5.25 \cdot (\text{inch})^2 = 2 \cdot 5.25 \cdot (\text{inch})^2 = 10.5 \text{ inch}^2$$

We will assume a continuous tiedown rod system. A conservative first-pass value for Δ_A can be:

$$\Delta_A = 0.25 \cdot \text{inch} = 0.25 \cdot \text{inch} = 0.25 \text{ inch}$$

North-South Direction

Shear Wall Rigidities With the deflections of each shear wall, along with the force demand of each shear wall, we can obtain the stiffnesses (and therefore, relative stiffnesses) of all the walls.

$$k = \frac{F_x}{\delta_{sw}}$$

		Direction	unit shear	wall length	wall height \
Level	Wall				
Roof	A-2	N-S	253.957181	30.75	10.0
	A-2.4	N-S	327.948065	30.75	10.0
	A-3.7	N-S	174.083810	20.00	10.0
	A-4	N-S	118.678605	20.00	10.0
	A-5	N-S	117.703076	30.75	10.0
	A-5.1	N-S	211.524369	30.75	10.0
	A-6	N-S	150.966989	30.75	10.0
	A-6.1	N-S	126.232285	30.75	10.0
	A-6.8	N-S	138.386407	30.75	10.0
	A-7	N-S	154.591903	30.75	10.0
	A-7.8	N-S	150.479225	20.00	10.0
	A-8	N-S	121.327990	30.75	10.0
	A-8.2	N-S	105.237105	20.00	10.0
	A-8.9	N-S	206.620074	30.75	10.0
	A-9	N-S	188.069046	30.75	10.0
Level 4	A-2	N-S	536.694237	30.75	10.5
	A-2.4	N-S	693.061071	30.75	10.5
	A-3.7	N-S	367.895788	20.00	10.5
	A-4	N-S	250.806545	20.00	10.5
	A-5	N-S	248.744936	30.75	10.5
	A-5.1	N-S	447.019885	30.75	10.5
	A-6	N-S	319.042418	30.75	10.5
	A-6.1	N-S	266.769931	30.75	10.5
	A-6.8	N-S	292.455550	30.75	10.5

	A-7	N-S	326.703041	30.75	10.5
	A-7.8	N-S	318.011613	20.00	10.5
	A-8	N-S	256.405559	30.75	10.5
	A-8.2	N-S	222.400279	20.00	10.5
	A-8.9	N-S	436.655512	30.75	10.5
	A-9	N-S	397.451147	30.75	10.5
Level 3	A-2	N-S	728.650626	30.75	10.5
	A-2.4	N-S	940.944301	30.75	10.5
	A-3.7	N-S	499.478992	20.00	10.5
	A-4	N-S	340.511102	20.00	10.5
	A-5	N-S	337.712129	30.75	10.5
	A-5.1	N-S	606.902956	30.75	10.5
	A-6	N-S	433.152513	30.75	10.5
	A-6.1	N-S	362.184022	30.75	10.5
	A-6.8	N-S	397.056470	30.75	10.5
	A-7	N-S	443.553068	30.75	10.5
	A-7.8	N-S	431.753027	20.00	10.5
	A-8	N-S	348.112684	30.75	10.5
	A-8.2	N-S	301.944927	20.00	10.5
	A-8.9	N-S	592.831618	30.75	10.5
	A-9	N-S	539.605249	30.75	10.5
Level 2	A-2	N-S	829.582306	30.75	11.5
	A-2.4	N-S	1071.282609	30.75	11.5
	A-3.7	N-S	568.666133	20.00	11.5
	A-4	N-S	387.678230	20.00	11.5
	A-5	N-S	384.491548	30.75	11.5
	A-5.1	N-S	690.970317	30.75	11.5
	A-6	N-S	493.152202	30.75	11.5
	A-6.1	N-S	412.353254	30.75	11.5
	A-6.8	N-S	452.056185	30.75	11.5
	A-7	N-S	504.993427	30.75	11.5
	A-7.8	N-S	491.558861	20.00	11.5
	A-8	N-S	396.332773	30.75	11.5
	A-8.2	N-S	343.769922	20.00	11.5
	A-8.9	N-S	674.949836	30.75	11.5
	A-9	N-S	614.350625	30.75	11.5

		shear stiffness	nail spacing	end post	youngs modulus \
Level	Wall				
Roof	A-2	14.0	6.0		1600000.0
	A-2.4	14.0	6.0		1600000.0
	A-3.7	14.0	6.0		1600000.0
	A-4	14.0	6.0		1600000.0
	A-5	14.0	6.0		1600000.0
	A-5.1	14.0	6.0		1600000.0
	A-6	14.0	6.0		1600000.0
	A-6.1	14.0	6.0		1600000.0

	A-6.8	14.0	6.0	1600000.0
	A-7	14.0	6.0	1600000.0
	A-7.8	14.0	6.0	1600000.0
	A-8	14.0	6.0	1600000.0
	A-8.2	14.0	6.0	1600000.0
	A-8.9	14.0	6.0	1600000.0
	A-9	14.0	6.0	1600000.0
Level 4	A-2	17.0	4.0	1600000.0
	A-2.4	19.0	3.0	1600000.0
	A-3.7	14.0	6.0	1600000.0
	A-4	14.0	6.0	1600000.0
	A-5	14.0	6.0	1600000.0
	A-5.1	17.0	4.0	1600000.0
	A-6	14.0	6.0	1600000.0
	A-6.1	14.0	6.0	1600000.0
	A-6.8	14.0	6.0	1600000.0
	A-7	14.0	6.0	1600000.0
	A-7.8	14.0	6.0	1600000.0
	A-8	14.0	6.0	1600000.0
	A-8.2	14.0	6.0	1600000.0
	A-8.9	14.0	6.0	1600000.0
	A-9	14.0	6.0	1600000.0
Level 3	A-2	19.0	3.0	1600000.0
	A-2.4	23.0	2.0	1600000.0
	A-3.7	17.0	4.0	1600000.0
	A-4	14.0	6.0	1600000.0
	A-5	14.0	6.0	1600000.0
	A-5.1	17.0	4.0	1600000.0
	A-6	14.0	6.0	1600000.0
	A-6.1	14.0	6.0	1600000.0
	A-6.8	14.0	6.0	1600000.0
	A-7	14.0	6.0	1600000.0
	A-7.8	14.0	6.0	1600000.0
	A-8	14.0	6.0	1600000.0
	A-8.2	14.0	6.0	1600000.0
	A-8.9	17.0	4.0	1600000.0
	A-9	17.0	4.0	1600000.0
Level 2	A-2	19.0	3.0	1600000.0
	A-2.4	23.0	2.0	1600000.0
	A-3.7	17.0	4.0	1600000.0
	A-4	14.0	6.0	1600000.0
	A-5	14.0	6.0	1600000.0
	A-5.1	19.0	3.0	1600000.0
	A-6	17.0	4.0	1600000.0
	A-6.1	14.0	6.0	1600000.0
	A-6.8	17.0	4.0	1600000.0
	A-7	17.0	4.0	1600000.0

A-7.8	17.0	4.0	1600000.0
A-8	14.0	6.0	1600000.0
A-8.2	14.0	6.0	1600000.0
A-8.9	19.0	3.0	1600000.0
A-9	17.0	4.0	1600000.0

		end post area	Delta_A	delta_sw	cumulative shear \
Level	Wall				
Roof	A-2	10.5	0.25	0.266632	7809.183325
	A-2.4	10.5	0.25	0.320628	10084.402984
	A-3.7	10.5	0.25	0.253490	3481.676193
	A-4	10.5	0.25	0.212596	2373.572094
	A-5	10.5	0.25	0.167197	3619.369601
	A-5.1	10.5	0.25	0.235665	6504.374356
	A-6	10.5	0.25	0.191472	4642.234923
	A-6.1	10.5	0.25	0.173422	3881.642761
	A-6.8	10.5	0.25	0.182291	4255.382013
	A-7	10.5	0.25	0.194118	4753.701016
	A-7.8	10.5	0.25	0.236068	3009.584505
	A-8	10.5	0.25	0.169843	3730.835694
	A-8.2	10.5	0.25	0.202675	2104.742105
	A-8.9	10.5	0.25	0.232086	6353.567289
	A-9	10.5	0.25	0.218548	5783.123167
Level 4	A-2	10.5	0.25	0.426475	16503.347775
	A-2.4	10.5	0.25	0.480798	21311.627941
	A-3.7	10.5	0.25	0.417312	7357.915758
	A-4	10.5	0.25	0.326268	5016.130894
	A-5	10.5	0.25	0.276384	7648.906777
	A-5.1	10.5	0.25	0.369480	13745.861455
	A-6	10.5	0.25	0.330367	9810.554345
	A-6.1	10.5	0.25	0.290226	8203.175384
	A-6.8	10.5	0.25	0.309950	8993.008149
	A-7	10.5	0.25	0.336250	10046.118503
	A-7.8	10.5	0.25	0.378524	6360.232266
	A-8	10.5	0.25	0.282267	7884.470935
	A-8.2	10.5	0.25	0.304180	4448.005572
	A-8.9	10.5	0.25	0.420685	13427.157006
	A-9	10.5	0.25	0.390579	12221.622785
Level 3	A-2	10.5	0.25	0.501104	22406.006745
	A-2.4	10.5	0.25	0.531796	28934.037258
	A-3.7	10.5	0.25	0.453519	9989.579834
	A-4	10.5	0.25	0.396019	6810.222033
	A-5	10.5	0.25	0.344704	10384.647963
	A-5.1	10.5	0.25	0.471098	18662.265904
	A-6	10.5	0.25	0.417995	13319.439778
	A-6.1	10.5	0.25	0.363497	11137.158685
	A-6.8	10.5	0.25	0.390276	12209.486463

	A-7	10.5	0.25	0.425982	13639.256835
	A-7.8	10.5	0.25	0.466965	8635.060534
	A-8	10.5	0.25	0.352691	10704.465019
	A-8.2	10.5	0.25	0.366031	6038.898543
	A-8.9	10.5	0.25	0.462154	18229.572239
	A-9	10.5	0.25	0.428325	16592.861418
Level 2	A-2	10.5	0.25	0.615150	25509.655919
	A-2.4	10.5	0.25	0.654368	32941.940221
	A-3.7	10.5	0.25	0.549028	11373.322664
	A-4	10.5	0.25	0.476238	7753.564603
	A-5	10.5	0.25	0.418384	11823.115086
	A-5.1	10.5	0.25	0.527989	21247.337256
	A-6	10.5	0.25	0.438714	15164.430219
	A-6.1	10.5	0.25	0.441926	12679.862556
	A-6.8	10.5	0.25	0.409946	13900.727701
	A-7	10.5	0.25	0.447003	15528.547894
	A-7.8	10.5	0.25	0.494075	9831.177218
	A-8	10.5	0.25	0.428389	12187.232761
	A-8.2	10.5	0.25	0.438581	6875.398447
	A-8.9	10.5	0.25	0.517915	20754.707461
	A-9	10.5	0.25	0.523555	18891.281713

		wall stiffness	total level stiffness \
Level	Wall		
Roof	A-2	29288.294664	328285.101560
	A-2.4	31452.035556	328285.101560
	A-3.7	13734.941315	328285.101560
	A-4	11164.701275	328285.101560
	A-5	21647.313323	328285.101560
	A-5.1	27600.052773	328285.101560
	A-6	24244.951226	328285.101560
	A-6.1	22382.701677	328285.101560
	A-6.8	23343.859376	328285.101560
	A-7	24488.769906	328285.101560
	A-7.8	12748.803355	328285.101560
	A-8	21966.439389	328285.101560
	A-8.2	10384.813322	328285.101560
	A-8.9	27375.885860	328285.101560
	A-9	26461.538542	328285.101560
Level 4	A-2	38697.132367	420325.638268
	A-2.4	44325.563615	420325.638268
	A-3.7	17631.691186	420325.638268
	A-4	15374.276740	420325.638268
	A-5	27674.949785	420325.638268
	A-5.1	37203.260574	420325.638268
	A-6	29695.919368	420325.638268
	A-6.1	28264.819587	420325.638268

	A-6.8	29014.353815	420325.638268
	A-7	29876.942671	420325.638268
	A-7.8	16802.722842	420325.638268
	A-8	27932.713296	420325.638268
	A-8.2	14622.933350	420325.638268
	A-8.9	31917.340188	420325.638268
	A-9	31291.018883	420325.638268
Level 3	A-2	44713.322701	477417.186205
	A-2.4	54408.199416	477417.186205
	A-3.7	22026.835054	477417.186205
	A-4	17196.719905	477417.186205
	A-5	30126.271281	477417.186205
	A-5.1	39614.442666	477417.186205
	A-6	31865.047545	477417.186205
	A-6.1	30638.955599	477417.186205
	A-6.8	31284.222309	477417.186205
	A-7	32018.376091	477417.186205
	A-7.8	18491.881022	477417.186205
	A-8	30350.837865	477417.186205
	A-8.2	16498.323034	477417.186205
	A-8.9	39444.787517	477417.186205
	A-9	38738.964198	477417.186205
Level 2	A-2	41469.006117	469393.643566
	A-2.4	50341.599805	469393.643566
	A-3.7	20715.375977	469393.643566
	A-4	16280.850199	469393.643566
	A-5	28259.014036	469393.643566
	A-5.1	40242.038416	469393.643566
	A-6	34565.667563	469393.643566
	A-6.1	28692.246158	469393.643566
	A-6.8	33908.720614	469393.643566
	A-7	34739.264924	469393.643566
	A-7.8	19898.143382	469393.643566
	A-8	28448.955948	469393.643566
	A-8.2	15676.469469	469393.643566
	A-8.9	40073.604071	469393.643566
	A-9	36082.686889	469393.643566

		total level cumulative shear	relative wall stiffness \
Level	Wall		
Roof	A-2	72387.392027	0.089216
	A-2.4	72387.392027	0.095807
	A-3.7	72387.392027	0.041838
	A-4	72387.392027	0.034009
	A-5	72387.392027	0.065941
	A-5.1	72387.392027	0.084073
	A-6	72387.392027	0.073853

	A-6.1	72387.392027	0.068181
	A-6.8	72387.392027	0.071108
	A-7	72387.392027	0.074596
	A-7.8	72387.392027	0.038835
	A-8	72387.392027	0.066913
	A-8.2	72387.392027	0.031634
	A-8.9	72387.392027	0.083391
	A-9	72387.392027	0.080605
Level 4	A-2	152978.135544	0.092065
	A-2.4	152978.135544	0.105455
	A-3.7	152978.135544	0.041948
	A-4	152978.135544	0.036577
	A-5	152978.135544	0.065842
	A-5.1	152978.135544	0.088511
	A-6	152978.135544	0.070650
	A-6.1	152978.135544	0.067245
	A-6.8	152978.135544	0.069028
	A-7	152978.135544	0.071080
	A-7.8	152978.135544	0.039975
	A-8	152978.135544	0.066455
	A-8.2	152978.135544	0.034790
	A-8.9	152978.135544	0.075935
	A-9	152978.135544	0.074445
Level 3	A-2	207692.959252	0.093657
	A-2.4	207692.959252	0.113964
	A-3.7	207692.959252	0.046137
	A-4	207692.959252	0.036020
	A-5	207692.959252	0.063103
	A-5.1	207692.959252	0.082977
	A-6	207692.959252	0.066745
	A-6.1	207692.959252	0.064176
	A-6.8	207692.959252	0.065528
	A-7	207692.959252	0.067066
	A-7.8	207692.959252	0.038733
	A-8	207692.959252	0.063573
	A-8.2	207692.959252	0.034557
	A-8.9	207692.959252	0.082621
	A-9	207692.959252	0.081143
Level 2	A-2	236462.301718	0.088346
	A-2.4	236462.301718	0.107248
	A-3.7	236462.301718	0.044132
	A-4	236462.301718	0.034685
	A-5	236462.301718	0.060203
	A-5.1	236462.301718	0.085732
	A-6	236462.301718	0.073639
	A-6.1	236462.301718	0.061126
	A-6.8	236462.301718	0.072239

A-7	236462.301718	0.074009
A-7.8	236462.301718	0.042391
A-8	236462.301718	0.060608
A-8.2	236462.301718	0.033397
A-8.9	236462.301718	0.085373
A-9	236462.301718	0.076871

direct shear force		
Level	Wall	
Roof	A-2	6458.116002
	A-2.4	6935.224343
	A-3.7	3028.576614
	A-4	2461.834559
	A-5	4773.267347
	A-5.1	6085.855954
	A-6	5346.050676
	A-6.1	4935.421660
	A-6.8	5147.358476
	A-7	5399.813086
	A-7.8	2811.131611
	A-8	4843.635157
	A-8.2	2289.868013
	A-8.9	6036.426790
	A-9	5834.811738
Level 4	A-2	14083.878359
	A-2.4	16132.354207
	A-3.7	6417.079994
	A-4	5595.490679
	A-5	10072.338763
	A-5.1	13540.181518
	A-6	10807.873621
	A-6.1	10287.022747
	A-6.8	10559.816834
	A-7	10873.757319
	A-7.8	6115.375743
	A-8	10166.152173
	A-8.2	5322.038145
	A-8.9	11616.363003
	A-9	11388.412441
Level 3	A-2	19451.839142
	A-2.4	23669.445237
	A-3.7	9582.433745
	A-4	7481.166891
	A-5	13105.968144
	A-5.1	17233.650284
	A-6	13862.395851
	A-6.1	13329.003523

	A-6.8	13609.716820
	A-7	13929.099062
	A-7.8	8044.606693
	A-8	13203.662361
	A-8.2	7177.340139
	A-8.9	17159.844436
	A-9	16852.786923
Level 2	A-2	20890.475981
	A-2.4	25360.144359
	A-3.7	10435.602509
	A-4	8201.660514
	A-5	14235.794615
	A-5.1	20272.377268
	A-6	17412.841917
	A-6.1	14454.040145
	A-6.8	17081.897539
	A-7	17500.293531
	A-7.8	10023.912442
	A-8	14331.479979
	A-8.2	7897.196957
	A-8.9	20187.526582
	A-9	18177.057382

East-West Direction

Shear Wall Rigidities With the deflections of each shear wall, along with the force demand of each shear wall, we can obtain the stiffnesses (and therefore, relative stiffnesses) of all the walls.

$$k = \frac{F_x}{\delta_{sw}}$$

[illegible]

Roof	A-B.1	14.0	6.0	1600000.0
	A-B.2.5	14.0	6.0	1600000.0
	A-B.3.5	14.0	6.0	1600000.0
	A-B.4	14.0	6.0	1600000.0
	A-B.4.8	14.0	6.0	1600000.0
...				
Level 2	A-C.6.5	19.0	3.0	1600000.0
	A-C.6.9	23.0	2.0	1600000.0
	A-C.7.5	19.0	3.0	1600000.0
	A-C.8.5	23.0	2.0	1600000.0
	A-C.9.2	17.0	4.0	1600000.0

		end post area	Delta_A	delta_sw	cumulative shear \
Level	Wall				
Roof	A-B.1	10.5	0.25	0.305430	5311.031480
	A-B.2.5	10.5	0.25	0.386928	3501.346680
	A-B.3.5	10.5	0.25	0.492407	2517.822331
	A-B.4	10.5	0.25	0.396192	2989.914019
	A-B.4.8	10.5	0.25	0.400676	2517.822331
...					
Level 2	A-C.6.5	10.5	0.25	0.729824	7967.751471
	A-C.6.9	10.5	0.25	0.957567	6343.358268
	A-C.7.5	10.5	0.25	1.456960	9766.921158
	A-C.8.5	10.5	0.25	0.890932	10964.654121
	A-C.9.2	10.5	0.25	0.520560	14562.993495

		wall stiffness	total level stiffness \
Level	Wall		
Roof	A-B.1	17388.715036	215842.289617
	A-B.2.5	9049.090153	215842.289617
	A-B.3.5	5113.295584	215842.289617
	A-B.4	7546.634903	215842.289617
	A-B.4.8	6283.928620	215842.289617
...			
Level 2	A-C.6.5	10917.365908	327888.366997
	A-C.6.9	6624.454117	327888.366997
	A-C.7.5	6703.631934	327888.366997
	A-C.8.5	12306.955097	327888.366997
	A-C.9.2	27975.637980	327888.366997

		total level cumulative shear	relative wall stiffness \
Level	Wall		
Roof	A-B.1	72387.392027	0.080562
	A-B.2.5	72387.392027	0.041925
	A-B.3.5	72387.392027	0.023690
	A-B.4	72387.392027	0.034964
	A-B.4.8	72387.392027	0.029114

...		...	...
Level 2	A-C.6.5	236462.301718	0.033296
	A-C.6.9	236462.301718	0.020203
	A-C.7.5	236462.301718	0.020445
	A-C.8.5	236462.301718	0.037534
	A-C.9.2	236462.301718	0.085321

direct shear force

Level	Wall	
Roof	A-B.1	5831.682635
	A-B.2.5	3034.808598
	A-B.3.5	1714.854548
	A-B.4	2530.927652
	A-B.4.8	2107.451720
...		...
Level 2	A-C.6.5	7873.245077
	A-C.6.9	4777.338344
	A-C.7.5	4834.438780
	A-C.8.5	8875.371078
	A-C.9.2	20175.109625

[72 rows x 16 columns]

Story Drift We can check the story drift at each shear wall line. This is conservative but all-encompassing.

The story drift is defined as:

$$\delta_x = \frac{C_d \delta_{xe}}{I_e} \quad (\text{ASCE 7-16 Eq. 12.8-15})$$

We can get the elastic story displacement, δ_{xe} , by getting a displacement for each shear line (which will be conservative). Since we got a stiffness using δ_{sw} , we can add those stiffnesses up for each line then get a displacement with $u = F/k$.

The story drift can't exceed the allowable story drift, Δ_a , which is defined in ASCE 7-16 Table 12.12-1. In our case:

- Structure is not masonry shear wall
- Four stories or less above the base
- Interior walls, partitions, ceilings, and exterior wall systems are designed to accommodate story drifts.

Therefore:

$$\Delta_a = 0.025h_{sx}$$

where h_{sx} is the story height below level x .

The ratio of the story drift, Δ_x , and the the maximum story drift, Δ_a , must not exceed 1

$$\text{ratio}_{\text{drift}} = \frac{\Delta_x}{\Delta_a} \leq 1$$

$$\text{ratio}_{\text{drift,max}} = 1.689$$

$$\text{check}_{\Delta} = 1.69 > 1 \quad \text{NG !}$$

Table 10: **SHEAR WALLS WITH INSUFFICIENT STIFFNESS**

Level	Wall	shear stiffness	story drift dcr
Level 4	A-C.7.5	23.0	1.13
	A-C.1.8	17.0	1.14
Level 3	A-C.6.9	19.0	1.11
	A-C.7.5	17.0	1.62
	A-B.6.5	17.0	1.23
	A-B.8.8	17.0	1.25
	A-C.1.8	17.0	1.29
Level 2	A-C.6.9	23.0	1.11
	A-C.7.5	19.0	1.69
	A-C.8.5	23.0	1.03